Predicting Spotify Track Popularity Using Audio Features and Genre Metadata

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Figure 1: Conceptual illustration of music streaming analytics. Source: Magnific.

Abstract

Music streaming platforms provide large-scale music metadata and audio features that can be used to study patterns in track popularity. In this project, we analyze a Spotify Tracks dataset containing 114,000 songs across 114 genres, with audio features, genre labels, explicit content indicators, and track popularity scores. We define a supervised regression task that predicts a track's numeric popularity score based on its audio characteristics and lightweight metadata. We first conduct exploratory data analysis to examine the distribution of popularity and its relationship with features such as danceability, energy, loudness, acousticness, valence, tempo, explicitness, and genre. We then compare simple baselines, including a global mean baseline and a genre mean baseline, with Ridge Regression, Random Forest, and histogram-based Gradient Boosting regressors. Model performance is evaluated using mean absolute error, root mean squared error, and R^2. Our best-performing model, HistGradientBoosting, achieves an RMSE of 17.78 and R^2 of 0.369 on a grouped test split. The results suggest that Spotify popularity is partially predictable from content-based features, but metadata, especially genre, contributes much more predictive signal than individual audio features alone.

Keywords

Spotify, popularity prediction, audio features, music information retrieval, regression, machine learning

1 Introduction

Music streaming platforms such as Spotify provide large collections of tracks together with structured metadata and machine-generated audio features. These features describe musical characteristics such as danceability, energy, acousticness, loudness, valence, tempo, and instrumentalness. At the same time, music popularity is influenced not only by acoustic content, but also by genre, artist recognition, playlist placement, marketing, release timing, and broader social trends. This makes popularity prediction an interesting data mining problem: the available features may contain useful signals, but they may not fully explain why some tracks become more popular than others.

The primary research question of this project is: **Can Spotify track popularity be predicted from audio features and lightweight metadata?** We model the problem as a supervised regression task, where the target variable is `popularity`, a numeric Spotify popularity score from 0 to 100. The predictors include audio features and metadata fields such as `track_genre`, `duration_ms`, and `explicit`. We intentionally exclude raw identifiers and high-cardinality text fields such as `track_id`, `artists`, `album_name`, and `track_name` from the main model input to reduce memorization and focus on generalizable patterns.

This project contributes an empirical comparison between simple baselines, linear regression, and nonlinear tree-based models. In addition to model comparison, we conduct an ablation study to test whether genre metadata improves over audio-only prediction, and we use error analysis to identify where the model fails. The results show that popularity is only partially predictable from these features, with the best model still making large errors for high-popularity tracks.

2 Dataset and Exploratory Data Analysis

2.1 Dataset Overview

We use the Spotify Tracks Dataset with audio features, which contains 114,000 rows and 20 original columns. Each row represents a track entry with metadata, audio features, a genre label, and a Spotify popularity score. The dataset includes 114 unique genres and 31,437 unique artists. The target variable, `popularity`, is an integer score ranging from 0 to 100.

Table 1: Dataset summary after feature engineering.

Property	Value
Rows	114,000
Original columns	20
Columns after engineering	23
Unique track IDs	89,741
Unique genres	114
Unique artists	31,437
Duplicated `track_id` rows	24,259

Only three fields contain missing values: `artists`, `album_name`, and `track_name`, each with one missing entry. Since these raw text fields are not directly used as model inputs, this missingness does not substantially affect the modeling pipeline. For simple engineered features, missing track names are filled with an empty string, and missing artist strings are handled similarly when computing artist counts.

2.2 Feature Design and Leakage Control

The raw dataset contains identifiers and high-cardinality text fields that could make a model memorize individual tracks or artists. We therefore exclude track ID, artist name, album name, and track name from the primary model input. Instead, we use Spotify audio features and lightweight metadata. The audio features include danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and time signature. The metadata features include duration, explicit status, and track genre. We also create three simple engineered features: track name length, artist count, and duration in minutes. These features summarize basic information without directly using the full text identity of a song or artist.

A duplicate-track check shows that the dataset has 89,741 unique track IDs but 114,000 total rows. Some tracks appear multiple times, often under multiple genres. This matters because if the same track appears in both the training and test sets, test performance could be overly optimistic. To avoid this leakage, our train/test split is grouped by `track_id`, ensuring that the same track never appears in both sets.

2.3 Popularity Distribution

The popularity score ranges from 0 to 100, with a mean of 33.24, a median of 35, and a standard deviation of 22.31. The 25th percentile is 17 and the 75th percentile is 50, indicating that many tracks are in the low-to-middle popularity range. This broad target distribution supports treating the task as regression rather than only predicting a small set of hit songs.

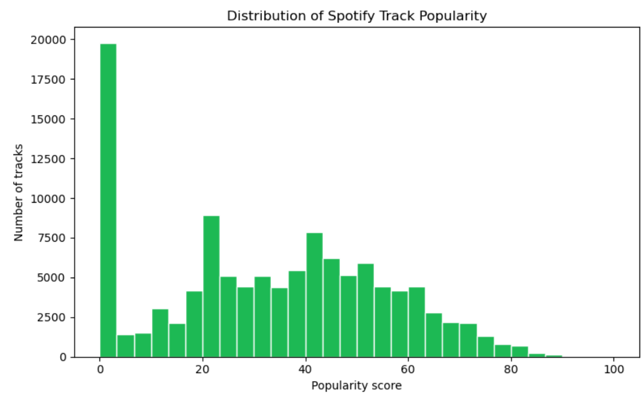

Figure 2: Distribution of Spotify track popularity scores.

2.4 Genre and Market Context

Popularity is not only an acoustic property. Genre can encode audience size, playlist ecosystems, and market context. In the dataset, genre-level average popularity varies substantially. For example, `pop-film` has an average popularity of 59.28 and `k-pop` has an average popularity of 56.90, while `iranian` has an average popularity of 2.21 and `romance` has an average popularity of 3.25. These differences suggest that `track_genre` may be a strong predictor even before considering audio features.

2.5 Audio Feature Distributions and Correlations

The main audio features have different distributional shapes. Features such as `danceability`, `energy`, and `valence` are spread across much of the 0 to 1 range, while `instrumentalness` and `speechiness` are highly skewed toward zero. This motivates using tree-based methods because they do not require normally distributed features and can capture threshold effects or nonlinear relationships.

Direct correlations between individual audio features and popularity are weak. The strongest absolute correlation is `instrumentalness` at -0.095, followed by `loudness` at 0.050, `speechiness` at -0.045, `valence` at -0.041, and `danceability` at 0.035. These small correlations suggest that popularity is unlikely to be explained by a single audio feature. Instead, the model must combine many weak signals and contextual metadata.

Overall, the EDA motivates three modeling decisions. First, we use regression because `popularity` is a broad numeric score. Second, we compare audio-only features against audio-plus-metadata features because genre-level patterns are strong. Third, we compare linear and nonlinear models because individual audio correlations are limited and feature distributions are skewed.

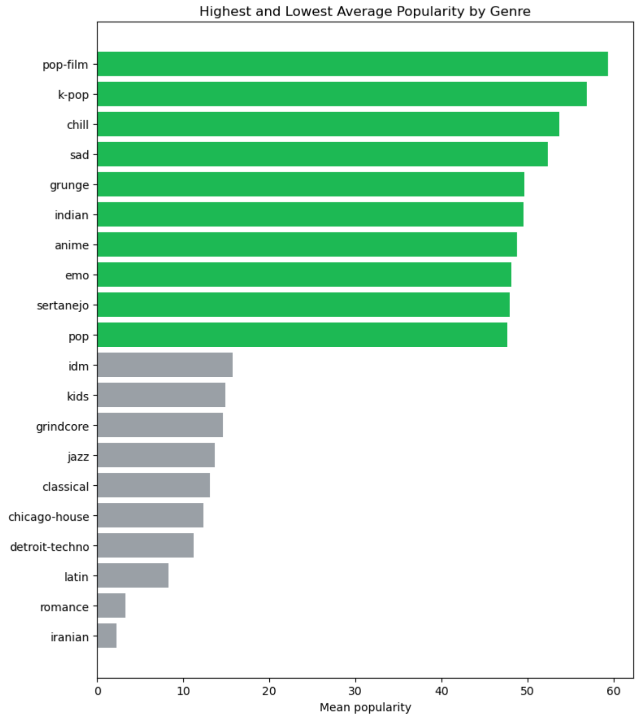

Figure 3: Highest and lowest average popularity by genre.

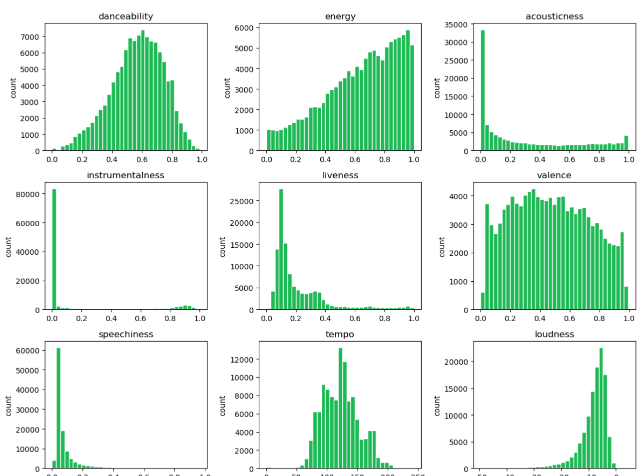

Figure 4: Distributions of selected Spotify audio features.

3 Predictive Task

The predictive task is to estimate a track's numeric Spotify popularity score from its audio features and lightweight metadata. Formally, for each track i, the model observes a feature vector x_i and predicts $\hat{y}_i$, where y_i is the true popularity score. Since popularity is bounded between 0 and 100, model predictions are clipped to the valid range $[0, 100]$ before evaluation.

We evaluate performance using mean absolute error (MAE), root mean squared error (RMSE), and R^2. MAE measures the average

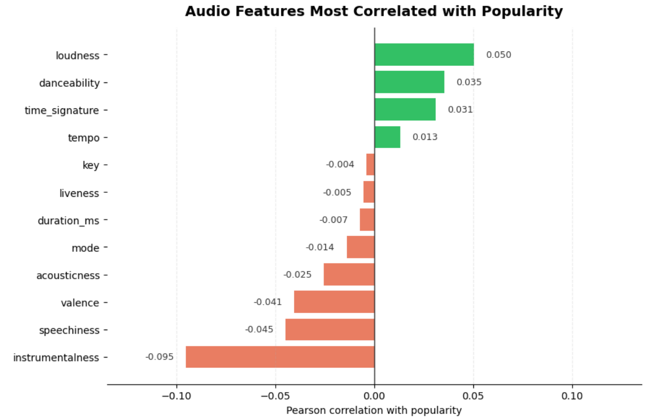

Figure 5: Audio features most correlated with Spotify popularity.

absolute prediction error, RMSE penalizes large errors more heavily, and R^2 measures the proportion of target variance explained by the model. The main evaluation metric for model selection is RMSE.

The train/test split uses `GroupShuffleSplit` with `track_id` as the grouping variable. This creates 91,150 training rows and 22,850 test rows, with zero overlapping track IDs between the two sets. This grouped split is important because the dataset contains repeated tracks; without grouping, a model could effectively see the same song during training and testing.

4 Related Work

This project is related to work in music information retrieval, hit song prediction, and content-based music recommendation. Prior work has studied whether audio content can be used to predict song popularity or commercial success. For example, Dhanaraj and Logan studied automatic hit song prediction using audio and lyric-based features [3]. Pachet and Roy later argued that hit song prediction is difficult because popularity is affected by many factors beyond the musical signal itself, including exposure, marketing, and social influence [4]. These findings are closely related to our project because our dataset contains Spotify audio features and genre metadata, but does not contain many external popularity drivers such as playlist placement, artist fame, social media attention, or release timing.

Spotify-style audio features are also commonly used in music information retrieval and recommendation tasks. Features such as danceability, energy, valence, acousticness, tempo, and instrumentalness provide a structured representation of musical content. In content-based recommendation, these features can be used to compare songs without relying on user listening histories. Large-scale music datasets, such as the Free Music Archive, have also been used to support research in music classification, recommendation, and analysis [5].

Compared with prior work, our project does not propose a new state-of-the-art algorithm. Instead, the main contribution is an empirical analysis of how well Spotify track popularity can be predicted from audio features and lightweight metadata. We compare simple baselines, a linear model, and nonlinear tree-based models, and we use ablation analysis to separate the value of audio-only features from genre and metadata features. Our conclusions are consistent with prior work: audio features contain some predictive signal, but popularity is only partially explained by musical content. The strong role of `track_genre` and the model's difficulty with high-popularity tracks suggest that market context and external exposure are important missing factors.

5 Model Design and Experimental Setup

We compare five models: two baselines and three machine learning regressors. The global mean baseline predicts the average popularity score from the training set for every test example. The genre mean baseline predicts the training-set mean popularity for each value of `track_genre`, falling back to the global mean for unseen genres. This is a strong baseline because EDA shows that genres differ substantially in average popularity.

The first learned model is Ridge Regression, a regularized linear model. Ridge Regression is useful as a linear baseline because it tests whether popularity can be approximated by a weighted combination of audio features and one-hot encoded genre variables. Numeric variables are median-imputed and standardized for Ridge Regression. Categorical variables are imputed with the most frequent value and one-hot encoded.

The second learned model is Random Forest Regression. Random forests can capture nonlinear feature interactions and are relatively robust to feature scaling. We use 120 trees, maximum depth 16, and minimum leaf size 8. The third model is HistGradientBoosting Regression, which is the main nonlinear model. Gradient boosting is commonly effective for tabular data because it builds an ensemble of weak learners sequentially and can capture nonlinear interactions between audio features, metadata, and engineered variables. We use 260 boosting iterations, learning rate 0.08, maximum 39 leaf nodes, and ℓ_2 regularization of 0.03.

To examine the value of different feature types, we run an ablation study with three feature groups: audio features only, audio plus metadata, and audio plus metadata plus simple engineered features. This allows us to directly test whether genre and lightweight engineered features improve prediction beyond audio features alone.

6 Results

6.1 Model Comparison

Table 2 reports the model comparison on the grouped test split. HistGradientBoosting performs best, with an MAE of 13.03, RMSE of 17.78, and R^2 of 0.369. It improves over the global mean baseline, which has an RMSE of 22.39 and approximately zero R^2. Ridge Regression and the genre mean baseline perform similarly, which suggests that genre is a major source of predictive signal. Random Forest performs worse than Ridge Regression and HistGradientBoosting in this run, possibly because the chosen random forest hyperparameters underfit or because gradient boosting is better suited to this tabular structure.

The improvement from the global mean baseline to HistGradientBoosting is meaningful but not complete. An RMSE of 17.78 on a 0–100 popularity scale indicates that the model captures useful structure but still leaves substantial unexplained variation. This supports the idea that content and genre features are informative, while many popularity drivers are missing from the dataset.

Table 2: Model performance on the grouped test split. Lower MAE/RMSE is better; higher R^2 is better.

Model	MAE	RMSE	R^2
HistGradientBoosting	13.030	17.782	0.369
Ridge Regression	14.167	19.270	0.259
Genre mean baseline	14.234	19.356	0.253
Random Forest	16.496	20.589	0.154
Global mean baseline	18.965	22.389	-0.000

6.2 Ablation Study

Table 3 shows the ablation study using HistGradientBoosting. Audio features alone achieve an RMSE of 20.58 and R^2 of 0.155. Adding metadata, especially `track_genre`, reduces RMSE to 18.07 and increases R^2 to 0.348. Adding simple engineered features further improves RMSE to 17.87 and R^2 to 0.363.

Table 3: Ablation study comparing feature groups.

Feature set	MAE	RMSE	R^2
Audio + metadata + engineered	13.134	17.867	0.363
Audio + metadata	13.266	18.072	0.348
Audio only	16.633	20.576	0.155

These results show that audio features alone contain some information, but metadata substantially improves prediction. This is consistent with the EDA finding that average popularity differs strongly by genre. The simple engineered features add a small additional benefit, but most of the gain comes from moving from audio-only features to audio plus metadata.

6.3 Error Analysis

The best model does not perform equally across popularity levels. Table 4 reports errors by true popularity bucket. The model has lower error for medium-popularity tracks, but much higher error for high-popularity tracks. For high-popularity tracks, the actual mean popularity is 73.78, while the predicted mean is only 39.78, giving a positive mean residual of 33.99. This means the model systematically underpredicts popular songs.

Table 4: Error analysis by true popularity bucket.

Bucket	Count	MAE	Actual Mean	Pred. Mean
Low	11,329	12.896	13.955	25.892
Medium	10,015	10.028	48.599	40.175
High	1,506	34.002	73.776	39.783

This result is important because it reveals a limitation of content-based popularity prediction. Many high-popularity songs may become popular due to external factors that are not included in the dataset, such as artist fame, playlist placement, marketing, social media trends, or recent release effects. The model therefore tends to regress predictions toward the middle of the popularity distribution.

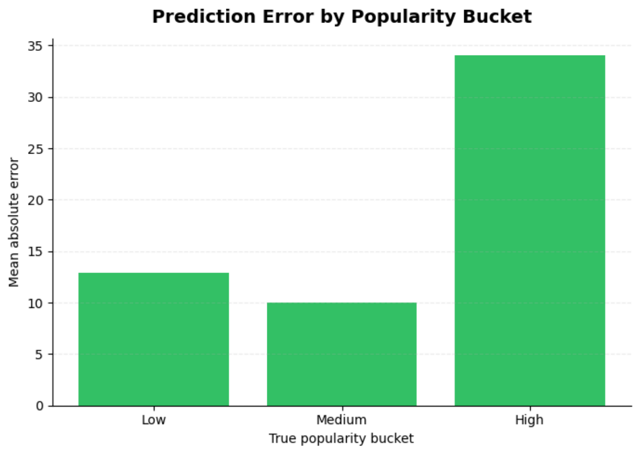

Figure 6: Prediction error by true popularity bucket.

6.4 Feature Importance

Permutation importance measures how much test RMSE increases when a feature is shuffled. The most important feature is `track_genre`, with an RMSE increase of 5.43 after permutation. This is much larger than the importance of any individual audio feature. The next most important features are `track_name_length`, `acousticness`, `danceability`, `valence`, `instrumentalness`, `duration_ms`, and `speechiness`. Most audio features individually have modest importance, which matches the weak direct correlations found in the EDA.

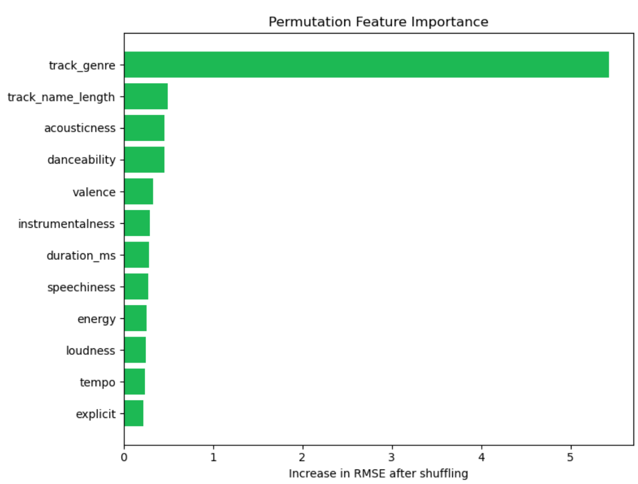

Figure 7: Permutation feature importance for the best model.

The feature importance results support the main conclusion that Spotify popularity is not purely an acoustic property. Genre appears to capture market and audience context, while audio features provide additional but weaker signals.

7 Discussion and Limitations

The results suggest that Spotify track popularity is partially predictable from audio features and lightweight metadata. The best model improves substantially over the global mean baseline and slightly outperforms both Ridge Regression and the genre mean baseline. However, the model still explains only part of the variation in popularity. The best R^2 is 0.369, meaning that a large share of popularity variation remains unexplained.

The strongest limitation is missing external context. Spotify popularity is time-dependent and may depend on playlists, artist popularity, release date, promotional campaigns, social media exposure, and user listening behavior. These variables are not included in the dataset. As a result, the model has particular difficulty with high-popularity tracks, which are likely influenced by factors beyond audio content.

A second limitation is the dataset granularity. The dataset contains repeated `track_id` values across multiple genre rows. We address this issue by using a grouped train/test split so that no track ID appears in both training and test sets, but the repeated rows still reflect the dataset's genre-label structure. Future work could aggregate the dataset to one row per unique track or use multi-label genre features.

A third limitation is that we avoid raw artist names and track names to reduce memorization. This makes the task more focused on content-based prediction, but it also removes information that is likely predictive in practice. Artist identity, album reputation, and text information may improve prediction, but they would need to be handled carefully to avoid leakage and overfitting.

Ethics and Privacy Statement

This project uses a public music dataset containing track-level metadata and audio features rather than private user listening histories. The main ethical risk is not personal privacy, but interpretation: model predictions should not be treated as objective judgments of musical quality. Popularity reflects platform exposure, audience behavior, and market context, so a model trained on historical popularity may reinforce existing visibility patterns if used for recommendation or promotion decisions.

8 Conclusion

This project examined whether Spotify track popularity can be predicted from audio features and genre metadata. We used a supervised regression framework, grouped train/test splitting by `track_id`, and compared global and genre baselines with Ridge Regression, Random Forest, and HistGradientBoosting. The best model was HistGradientBoosting, achieving an RMSE of 17.78 and R^2 of 0.369. The ablation study showed that audio features alone are useful but limited, while adding genre and lightweight metadata substantially improves performance. Error analysis showed that high-popularity tracks are often underpredicted, suggesting that popularity depends on external factors not captured by audio features. Overall, the project supports the conclusion that Spotify popularity is partially predictable from content-based features, but genre and broader market context play a central role.

References

[1] Sai Chaitanya Reddy. Spotify Tracks Dataset: Audio Features. Kaggle. Retrieved 2026 from https://www.kaggle.com/datasets/saichaitanyareddyai/spotify-tracks-dataset-audio-features.

[2] Spotify for Developers. Web API Reference: Audio Features. Retrieved 2026 from https://developer.spotify.com/documentation/web-api/reference/get-audio-features.

[3] R. Dhanaraj and B. Logan. 2005. Automatic Prediction of Hit Songs. In *Proceedings of the 6th International Conference on Music Information Retrieval (ISMIR)*.

[4] F. Pachet and P. Roy. 2008. Hit Song Science Is Not Yet a Science. In *Proceedings of the International Conference on Music Information Retrieval (ISMIR)*.

[5] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson. 2017. FMA: A Dataset for Music Analysis. In *Proceedings of the International Society for Music Information Retrieval Conference*.

[6] F. Pedregosa et al. 2011. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12, 2825–2830.